

AARUSHI SINGH

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EDUCATION

- Expected 12/25 **NEW YORK UNIVERSITY** New York, NY
The Courant Institute of Mathematical Sciences
M.S. in Mathematics in Finance
- **Expected Coursework:** penalized regression, decision trees, linear regression, Fama-French, Black-Scholes, stochastic processes, alternative data, object-oriented programming (Python)
- 08/20 - 12/23 **PURDUE UNIVERSITY** West Lafayette, IN
B.S. in Computer Science
- **Coursework:** operating systems (x86 Assembly, C), computer architecture (x86 Assembly, C), data structures and algorithms (Java), systems programming (C/C++, Bash Script), data mining and machine learning (Python), marketing management for new ventures
 - **Honors:** FactSet Research Systems Scholarship 2021, Entrepreneurship & Innovation Certificate

EXPERIENCE

- 10/23 - 08/24 **EXCELLENT REALM** Cupertino, CA
Software Engineer (Python, OpenAI, LangChain, React)
- Acquired 80 student clients in first month by implementing data-driven marketing strategy with Pandas for targeted school advertising in local area
 - Developed AI chatbot to teach computer science fundamentals for this tutoring startup
 - Integrated progress-tracking system into chatbot with React and Azure to monitor client performance and provide personalized learning experiences
- 06/22 - 08/22 **PUBLICIS SAPIENT** Boston, MA
Financial Services Software Engineer Intern (Python, PostgreSQL, React Native, Spring Boot)
- Built database to manage user transactions for stock trading app, ensuring data integrity and fast query performance for high-frequency data
 - Conducted market analysis of popular trading platforms using Python to track features and performance, providing insights for strategic positioning in competitive market
 - Facilitated front-end and back-end integration to create responsive and user-friendly interface
- 08/21 - 05/22 **THE DATA MINE** West Lafayette, IN
Data Science Researcher (R, SQL)
- Collaborated with CliftonLarsonAllen financial services firm to determine dynamic pricing model that maximizes profitability for their services while staying competitive
 - Analyzed CliftonLarsonAllen's employee data (e.g., billable hours, region) to create model
 - Trained in R to manipulate and aggregate dataframes, providing detailed analysis of pricing structures and identifying improvements through predictive insights on workforce utilization

PROJECT

- 10/24 - 12/24 **NEW YORK UNIVERSITY** New York, NY
Customer Spending Alternative Data for Trading (Python)
- Performed k-means clustering using Euclidean distance on credit card transaction data to uncover spending patterns among users, ensuring optimal cluster weights by preprocessing data
 - Created a constrained regression model to map spending clusters to quarterly company revenue data, achieving a Mean Absolute Percentage Error (MAPE) of 32.16% by leveraging L-BFGS-B optimization techniques
 - Applied insights to design a market-neutral portfolio by comparing predictions with Bloomberg estimates, identifying high-impact segments, and assessing stock price sensitivities.

COMPUTATIONAL SKILLS / OTHER

Programming Languages: Python, Java, C/C++, R, SQL, LaTeX, x86 Assembly, HTML, CSS, Bash Script, Javascript
Technologies: React, PostgreSQL, Django, Excel, OpenAI, Node, Spring Boot, Android Studio, Linux, Figma
Activities: Marketing Director for NYU Student Council; Recruitment Leader for Delta Kappa Delta; Treasurer for Association of South Asians at Purdue; Alumni Fundraising Cold Caller (Top 10 of 200 Students) for Rally Line